Ministry of Education
Government of India

AICTE

MoE's INNOVATION CELL
(GOVERNMENT OF INDIA)

SMART INDIA HACKATHON 2025

GOVERNMENT OF KERALA

Krishi Sakha

"Anywhere - Anytime"

Revolutionizing Indian Farming through Precision AI & Localized Language Models (LLMs)

Problem Statement

AI-Based Farmer Query Support and Advisory System

Krishi Sakha

"The Crisis of Information & Livelihood for Smallholder Farmers"

My village has no internet, how can I get advice?

✦ Offline-First AI Chat: An on-device AI, powered by llama.cpp, provides instant advisory with zero network required.

Who can I trust for a second opinion or local advice?

✦Community & Expert Connect: A trusted ecosystem where farmers can ask public questions to the community or connect privately with verified agricultural experts.

I can't describe this pest in words. How do I get help?

✦Multimodal AI Assistant: Ask questions using text, voice, or even by sending an image of the crop. Powered by Gemma for offline use, with a Gemini API backup for complex queries.

Am I getting the right price for my crops? Will it rain next week?

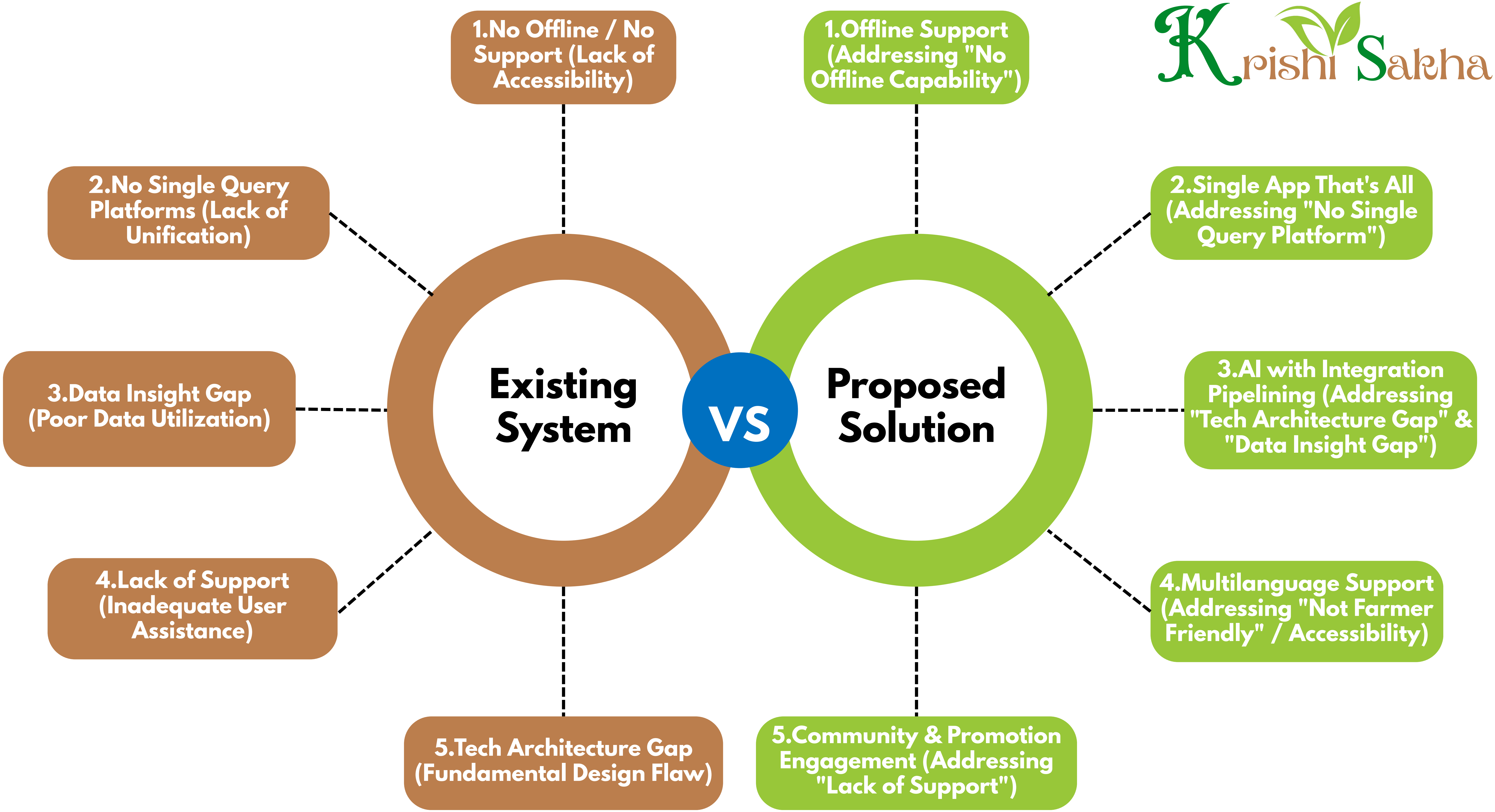
✦ Real-Time Intelligence Hub: Integrated weather forecasts from OpenMeteo and live mandi prices help in making profitable decisions.

I don't know which government schemes I am eligible for?

✦Proactive Scheme Finder: The app automatically finds personalized government schemes using the MyScheme API and sends push notifications for new benefits.

I can't read English / Hindi or Malyalam very well.

✦Inclusive & Voice-First UI: Designed for all literacy levels with a focus on icons over text and voice commands. Multilingual support is powered by leading Indian NLP models from AI4Bharat and is extendable with Bhashini APIs for full regional coverage.



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Existing System
VS
Proposed Solution
1.No Offline / No Support (Lack of Accessibility)
2.No Single Query Platforms (Lack of Unification)
3.Data Insight Gap (Poor Data Utilization)
4.Lack of Support (Inadequate User Assistance)
5.Tech Architecture Gap (Fundamental Design Flaw)
1.Offline Support (Addressing "No Offline Capability")
2.Single App That's All (Addressing "No Single Query Platform")
3.AI with Integration Pipelining (Addressing "Tech Architecture Gap" & "Data Insight Gap")
4.Multilanguage Support (Addressing "Not Farmer Friendly" / Accessibility)
5.Community & Promotion Engagement (Addressing "Lack of Support")

Key Feature 1: Community Support

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⭐ **Key Highlights**

📝 Farmers share posts, images & queries

✔ Posts are verified by Governing Agency before they are public.

👍 Suggestion/Solution-based points system for farmers.

🏆 Leaderboard showcasing top contributors

👨‍🌾 Private chat with Agro Experts / Agronomists

🌐 Safe, verified, knowledge-sharing platform

⚙ **App Architecture Summary**

Backend & Database (Supabase) Database (RSQL):
🗄Stores Users, Posts, Likes, Comments, Points, and Leaderboard data.

🪣Buckets (Image Storage): Securely stores post images.

⚙RPC Functions (Optimized Server Logic):
Fetches structured JSON responses.
Handles Scoring, Leaderboard logic, and Verified Post flows.
Enables faster, optimized execution.

Frontend + App Layer
📱Mobile UI: Integrated with the Supabase API.
🔐Authentication & Roles: Manages access for Farmers, Experts, and Governing Agency.

Key Feature 2: CROP ADVISORY SUPPORT

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Smart Planning & Advisory (The "Brain")

- Focuses on decision-making, helps farmer take decisions precisely.
- AI-Based Context Support Crop Suggestions.
- Soil Nutrition Guidance.
- Crop Advisory (Currently Kerala).

Protection & Diagnostics (The "Shield")

- Focuses on saving the crop from threats.
- Pest & Disease Diagnosis
- Weather-Linked Alerts
- Fertilizer & Pesticide Dosage

Accessibility & Inclusion (The "Reach")

- Focuses on how the user interacts with the app.
- Local Language Support: Easy communication for all farmers, including illiterate users through voice & audio.
- Expert Bridge: Directly bridges the gap between agriculture experts & farmers.
- Data-Driven Decisions: Improves decision-making using real-time field data.

Impact & Benefits (The "Result")

- Reduces Crop Losses: By preventing diseases early using TensorFlow Models.
- Increases Productivity: Achieved through accurate input usage.
- Saves Input Costs: Optimizes fertilizer and water consumption.
- Sustainable & Profitable: Promotes eco-friendly farming & better profit margins.

Key Feature 3: WEATHER AND MARKET PRICE

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Weather Updates & Forecasts

- Real-time weather updates for temperature, rainfall, humidity & wind
- 7–15 days forcast: Plan irrigation, sowing & spraying.
- Automatic alerts: Storms, heavy rain, frost & heat waves

Weather Based Advisory

– When to irrigate?

– When to spray pesticides?

– When to harvest or store?

Prevents crop loss & reduces unnecessary field visits.

Mandi/Market Price Updates

- Daily mandi/market price for major crops
- Price comparison across markets.
- Tracks trends & predicts price rise/fall.
- Notifies best selling time.

BENEFITS AND EMPOWERMENT

✓ Saves crops, Reduces wastage of water, fertilizers & labour, Increases selling profit.

✓ Eliminates dependency on middlemen, Empowers data-driven decision-making.

Key Feature 4: Government Scheme Awareness

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- **Scheme Awareness: Information about central & state programs**
- **Eligibility Check: Helps farmers know if they qualify**
- **Application Support: Step-by-step guidance in local language**
- **Real-Time Updates: New scheme alerts, deadlines, subsidy changes**
- **Document Support: Required documents & submission process**
- **Benefits Tracking: Scheme status and approval updates**

SUBSIDY AWARENESS

Suggest schemes for micro-irrigation, seeds, machinery, etc,.

MARKET LINKAGE

e-NAM(National Agriculture Market) integration in Krishi Sakha*.

WEATHER ALERTS

Ingests and integrates real-time user data with IMD reports and generates Weather Alerts, and employs a push notification to end-users.

TARGET IMPACT AND BENEFITS (Farmers)

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Impacts

- Encourages digital and smart farming.
- Reduces crop losses with timely guidance.
- Improves market linkage and selling opportunities.
- Connects farmers with government schemes easily.
- Enhances decision-making using AI insights.

Benefits

- Personalized crop advisory and recommendations.
- Access to weather, market rates, schemes in one app.
- Works offline for remote village support.
- Increases productivity and profit.
- Promotes sustainable & efficient farming.

Feasibility & Viability

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1. Feasibility (Technical & Adoption)

- "The system is technically feasible due to the hybrid LLM/RAG architecture, which allows rapid knowledge updates.”
- LLM/RAG Architecture, Rapid Update.
- We ensure high adoption through Multimodal Input and extreme Localization in vernacular languages.
- High Adoption, Extreme Localization.
- “The Offline Capability makes it usable for 70% of farmers with poor connectivity, guaranteeing access.”
- Offline Capability, Guaranteed Access.

2. Viability (Sustainability & Growth)

- “Krishi Sakha is highly scalable and sustainable.”
- The automated first line support dramatically reduces operational costs for government partners.
- High Scalability, Reduced OpEx.
- "We guarantee long-term reliability through the Learning Loop. where expert feedback continually improves the Al's accuracy."
- Long-Term Reliability, Learning Loop.
- "Future viability includes integration with Agri-Input Suppliers and expansion into Agri-Data Analytics services."

RESEARCH AND REFERENCES

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Feature	Krishi-Sakha	Govt Solutions	Existing Apps
Ai Assistant (Voice/Text/Image)	✓	✗	✗ / ✓
Offline-First Support	✓	✗	✗
Govt Scheme + Auto Notify	✓	✓	✗
Weather Forecasts	✓	✓	✓
Market Prices (Govt. Mandi Data)	✓	✓	✓
Offline Disease Detection	✓	✗	✗
SOS Emergency Alerts	✓	✗	✗
Multilingual Support	✓	✗ / ✓	✓
Farmer-Friendly UI (Voice + Icons)	✓	✗	✗
Integration with Govt Systems	✓ (AgriStack-ready)	✗	✗
Future Scalability (Automation, Data Sync, Health, Education, Disaster Management)	✓	✗	✗

Data Security

The security architecture of Krishi Sakha is built upon robust authentication and granular access control to ensure user data safety and system integrity.

I. User Data Safety & Authentication

Primary Authentication: The system uses Supabase Authentication.

Access Control: Access and operations are secured using Supabase & JWT Authentication integrated with User Roles.

Write Operations: All data write operations are strictly controlled and only allowed after backend and authentication.

II. Granular Security & Integrity Row-Level Security (RLS):

Implements Row level security for each table to restrict data access based on the user's identity and role, ensuring data separation and privacy.

User Role Management: User roles need to be added to the Login Page to tailor permissions and access levels.

Farmer Validation: Validation of the farmer is planned for future implementation.

III. Data Integration Security

Reverse Engineering: Key data sources— My Scheme, IMD (Indian Meteorological Department), and e-NAM— were Reverse Engineered.

Integration Purpose: This reverse engineering was done specifically for APIs and integrating them into the mobile application Krishi Sakha.

Future Implementation

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1.Offline AI Data Sync Model:

Goal: Ensure the AI's core functionality remains available and reliable during connectivity loss.

Implementation: Refine the existing offline model to allow for the caching of necessary AI weights and localized advisory knowledge.

Process: Optimize data synchronization to transfer minimal, actionable data (e.g., updated pest advisories or scheme changes) when the device briefly connects to the internet.

2. Community Feature: Knowledge Sharing & Connection Community Feature for Expert Advice and Connect:

Goal: Leverage peer-to-peer and expert knowledge sharing to accelerate learning and foster collective bargaining power.

Features: Create a dedicated platform for farmers to exchange information, discuss problems, and share best practices.

Function: Enable farmers to connect with local agri officers or high-ranking community members for verified advice, extending the existing Escalation System.

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Thank You!